

Financial development and economic growth in Vietnam

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Abstract By making use of a panel dataset that covers 61 provinces of Vietnam over the period 1997 to 2006, this paper examines the link between financial development and economic growth. Our analysis, which is based on endogenous growth theory, reveals that financial development has contributed to economic growth in Vietnam. We find that high ratio of credit to Gross Provincial Product (GPP) has accelerated economic growth in Vietnam. We also found a strong positive link between financial development and economic growth when alternative measures of financial development were used. The impact of foreign direct investment on economic growth will be stronger if more resources are invested in financial market development.

Keywords Financial Development · Economic Growth · Globalization · Vietnam

JEL Classification F20 · O11 · O16

1 Introduction

Financial development has contributed to impressive economic growth in a number of developing countries. In addition, it has been suggested that countries which are relatively more financially developed are better able to avoid or withstand currency

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crises (Federici and Carioli 2009). Therefore, enhancing the financial development of countries with developing economies may have important positive consequences for the many organisations and individuals within such countries that are affected by economic downturns. These issues are particularly salient given the recent global financial crisis.

Generally speaking, financial development not only increases the supply of capital but, given the appropriate host-country policies, it can also facilitate technological innovation. Technological innovation contributes to human capital formation which can further enhance prospects of economic growth (in fact, it can be argued that a bi-directional causality exists between technological innovation and human capital formation). In other words, financial development can facilitate economic growth through direct as well as indirect channels.¹

Endogenous growth literature [for example the work of Romer (1986), Lucas (1988, 1993)] highlights the role of financial development for long-run economic growth in developing countries. This literature suggests that financial development enhances economic growth through the impact of financial sector services on capital accumulation and technological innovation. These services include mobilization of savings, acquiring information about investments and allocation of resources, monitoring of managers and exerting corporate control and facilitation of risk reduction (Roubini and Sala-i-Martin 1992; King and Levine 1993).

On the other hand, theories of financial structure (which include the bank-based, the market-based, the financial services based and the law and finance based theories) provide alternative tools to analyse the relationship between financial structure and economic growth. For example, the bank-based theory emphasizes the positive role of commercial banks in economic development. It argues that banks can finance economic development effectively in the early stages of economic development because these banks that are unhampered by regulatory restrictions, can exploit economies of scale and scope in information gathering and processing. They can also be efficient in mobilizing resources and managing risks (Levine 2002; Beck and Levine 2004). In contrast, the market-based theory highlights the advantages of well-functioning markets in promoting successful economic performance. According to this theory, big, liquid and well-functioning markets foster growth and profit incentives, enhance corporate governance, facilitate risk management and diversification as well as customization of risk management devices (Levine 2002). The financial-services theory that is based on both the bank-based and the market-based views stresses the importance of the key financial services provided by financial systems. This theory suggests that financial services are crucial to new growth creation, industrial expansion and economic growth (Merton and Bodie 1995). Finally, the law and finance theory emphasizes the role of the legal system in creating a growth-promoting financial sector with legal rights and enforcement mechanisms, facilitating both financial markets and intermediaries. It also suggests that the legal system is critical to firm, industry and national economic success (La-Porta et al. 1998; Levine 1999).

Recent empirical studies have used endogenous growth models to investigate the impact of financial development on economic growth in host developing countries.

¹ See Allen and Gale (2000), Levine (2002), Beck and Levine (2004) and Luintel et al. (2008) for an interesting survey.

King and Levine (1993) used cross-country data to examine the impact of financial development on economic growth. They found that (i) various financial indicators are robustly positively correlated with economic growth, and (ii) government intervention in the financial system has a negative effect on the growth rate. By making use of panel data for the period 1986–1993 involving 44 countries, Demircuc-Kent and Levine (1996) found a positive relationship between economic growth and stock market development. By using time series data for 16 countries, Demetriades and Hussein (1996) found that finance is a leading sector in the process of economic development. Similarly, using time series data for 71 developing countries, Odedokun (1996) showed that financial intermediation promoted economic growth in approximately 85% of the countries. Hsu and Lin (2000) examined the relationship between economic growth and financial development by using Taiwanese data for the period 1964–1996. They found that both banking and stock market development are positively related to short-run and long-term economic growth. In addition, the financial depth as measured by the ratio of the broad monetary aggregate (M2) to GDP has strong effect on economic growth.

Using the generalized method of moments (GMM) and principal component analysis, Liu and Hsu (2006) investigated the impact of financial development on economic growth in three Asian countries, namely, Taiwan, Korea, and Japan. They found that (i) high investment promotes economic growth in Japan while high investment does not lead to better economic growth if investment is not allocated efficiently in both Taiwan and Korea; (ii) the finance aggregate has positive effects on Taiwan's economy, but has negative effects in the other two countries, (iii) the stock market development has positive effects on Taiwan's economic growth but its economy suffered from the Asian financial crisis in the late 1990s; (iv) capital outflows have a negative impact on all three countries while the effect of capital inflows is negative. Braun and Raddatz (2007), using a cross-country panel data involving 33 countries for the period 1970–2003, found that domestic financial development has a smaller effect on economic growth in countries that are open to trade and capital flows than among countries that are closed in both dimensions. Abu-Bader and Abu-Qarn (2008) employed four different measures of financial development to investigate the relationship between financial development and economic growth in Egypt during the period 1960–2001. They suggested the need to accelerate financial reforms that the Egyptian government launched in 1991 and to improve the efficiency of the financial system to stimulate saving and consequently long-term economic growth. In a recent study involving developed as well developing countries, Federici and Carioli (2009) have argued that relatively more financially developed countries are able to avoid currency crises.

This paper examines the role of financial development on economic growth within a developing country; in particular, this paper focuses on Vietnam. The introduction of the reform policy known as *Doi Moi* in 1986 marked the beginning of Vietnam's impressive economic growth. Due to lack of data, as indicated by Nguyen et al. (2008), studies on Vietnam are hard to find. By making use of a recently released panel dataset that provides comprehensive data on 61 Vietnamese provinces for the period 1997–2006, this paper attempts to empirically examine the relationship between financial development and economic growth in Vietnam. While the sample size utilized in this study is sufficiently large (610 observations), the

sample period covers only 11 years. It can be argued that the sample period is too short for an empirical study of long-run growth. However, due to the lack of data we have little choice but to utilize the data for the period 1997–2006.

The rest of this paper is organized as follows. Section 2 contains an overview of Vietnam's financial system. An empirical model based on endogenous growth theory is specified in Section 3. Section 4 contains empirical analysis, while Section 5 contains some concluding remarks.

2 Financial development in Vietnam

Prior to the introduction of economic reforms, under the centrally planned economy, Vietnam's single-tier banking system was dominated by the government. The State Bank of Vietnam provided nearly all domestic banking services through a vast branch network. Trade and infrastructure finance were managed by two specialized banks, namely, the bank for foreign trade of Vietnam (VCB) established in 1963 and the bank for investment and development of Vietnam (BIDV) established in 1958. These two banks were fully owned by the government. They were responsible for the financing of foreign trade, foreign exchange transactions, public expenditure management, infrastructure projects and the purchases of equipment for state owned industrial and agricultural enterprises.

The “*Doi Moi*” policy of 1986 reflected the realization that improving the efficiency and competitiveness of the banking sector was essential to mobilizing private and foreign resources needed to fuel further economic development. Accordingly, the Prime Minister enacted Decision No. 218/CT in 1987 and Decree No. 53/ND in 1988. As a result, the monobank system was split into a two-tier banking system consisting of the State Bank of Vietnam (SBV) as the central bank, and four specialized state commercial banks, which were later named as the four State Owned Commercial Banks (SOCBs).²

Since 1990 the Vietnamese financial system has made steady progress towards what is required for a market-based economy. The banking sector has grown significantly in terms of both structure and services. Starting from only four SOCBs and a small number of credit cooperatives in 1990, Vietnam's banking system has expanded to include more diversified banking institutions. In early 1994 the number of banks had risen to 60, including SOCBs, shareholding and joint venture banks, a large number of credit cooperatives and people's credit funds. Foreign banks have also been allowed to enter and operate in Vietnam and by September 1995 the number of foreign bank branches and representatives offices had increased to about 20 and 60 respectively (Harvie and Hoa 1997). The reform in the financial system resulted in the establishment of a large number of banks and branches during the mid 1990s, many of which were designed to promote domestic investment by providing services to both foreign and domestic entities operating in the country. This in turn led to a marked increase in bank credit from an annual average of 18% of GDP in

² The four SOCBs are the Bank for Foreign Trade of Vietnam (VCB), Vietnam Industrial and Commercial Bank (ICB), Bank for Investment and Development of Vietnam (BIDV), and Vietnam Bank for Agriculture and Rural Development (VBARD).

1991 to approximately 21% in 1997. Although bank credit was still largely channelled to the state sector, credit to the private sector rose gradually from an annual average of 1% of GDP in 1991 to about 10% in 1997 (see Table 1).

The significant increase in private, foreign, and mixed-ownership banks has marked improvements in financial services. The relatively large network of branches of commercial banks among provinces and cities has allowed them to promote the process of savings mobilization. As a result, the gross domestic saving rate increased sharply from 2.9% in 1990 to 16.2% in 1992, and to 18.0% in 1997 (ADB 1998). However, the saving rate was relatively lower than that of China, and other newly industrialized economies, such as Malaysia and Thailand (see ADB 1998). Although numerous new banks have entered the banking sector, most are small and the sector has remained dominated by the SOCBs. By the end of 1993, the SBVN accounted for almost 45% of total financial assets in the system, the SOCBs accounted for 49.5% and the other banks only 5.5% (Harvie and Hoa 1997). Consequently, the banking system remained lacking in both competition and innovation and hence Vietnam was regarded to be a substantially “underbanked country”.

In addition, Vietnam was far from being a financially independent country. The ratio of M2 to GDP at only 28% during the period 1990–1997 was far below the 120% in China and 90% in Thailand. However during this period, Vietnam attracted significant FDI which could be attributed to promulgation of a liberal foreign investment law in 1987. FDI inflow into Vietnam increased from US\$0.32 billion in 1988 to approximately US\$9.0 billion in 1996 with an annual growth rate of 28% (GSO 2006). In the early stages, the contribution of FDI to employment growth was small but there was a large increase in industrial output. In recent years, FDI inflows have played an important role, not only in providing investment capital, but also in stimulating export growth (see Xuan and Xing 2008).

Continuing the initial steps of restructuring the banking system that began in mid 1988, the government’s objectives for further reform were to develop a sound and competitive banking system that would help to maintain macroeconomic stability by encouraging financial discipline and efficient utilisation of savings. With these objectives in mind, the Law of the State Bank of Vietnam and the Law of Credit Institutions were enacted in 1997. Based on these laws, the SBV mainly implements monetary control through three policy instruments [(i) setting interest

Table 1 Distribution of credit in Vietnam (1991–1997)

	1991	1992	1993	1994	1995	1996	1997
Total Credit (Billions of VND)	14112	17122	27112	37951	47055	55323	66807
Credit to Private Enterprises (Billions of VND)	1026	2770	7698	12936	18292	24085	30979
Credit to State Owned Enterprises (Billions of VND)	9129	12439	15511	20464	24079	26809	31429
Total Credit (% of GDP)	18	15	19	21	21	20	21
Credit to Private Enterprises (% of GDP)	1	3	5	7	8	9	10
Credit to State Owned Enterprises (% of GDP)	12	11	11	11	11	10	10

(ADB 2008; GSO 2008)

rates on banking system liabilities and assets; (ii) varying the cost and availability of central bank credit to banks; and (iii) setting reserve requirements]. This means that the SBV has gradually shifted its role to that of a modern central bank and has gradually adopted credit and money controlling policies along the lines of central banks in market-based economies.

It is interesting to note that the Vietnamese economy has expanded since 2000 by an annual average growth rate of 7% [from 6.8% in 2000 to 7.7% in 2004, and to 8.5% in 2007]. The gross domestic saving rate also rose steadily from 31.7% in 2000 to 32.0% in 2004 and to over 35.0% in 2007. In addition, the annual average growth rate of M2 was approximately 25% during 2000–2007 (IMF 2008). Although the banking sector has grown dramatically since 1990, it still has many problems. The main problem faced by the banking sector is the high percentage of nonperforming loans. As of the end of 1997, the total overdue loans were estimated at US\$610 million. Of this total, almost 75% was held by SOCBs and 33% was owed by SOEs. The private sector's share of overdue loans rose to 67% in 1997 from 41% in 1994 (ADB 2003). However, the problem has improved since 2000. The ratio of overdue loans to total loans has declined from 9.7% in 2000 to 8.5% in 2002 and 3.2% in 2006 (IMF 2008). It is perhaps also worth mentioning that the Vietnamese financial system consists primarily of the banking sector. The relationship between financial development and economic growth in Vietnam is investigated in the next section.

3 Methodology and data

For the purposes of the present study, we make use of a recently released panel dataset which provides annual data on 61 provinces of Vietnam for the period 1997–2006. Unfortunately, comparable provincial data on relevant variables is not available for previous years. The choice of the determinants of economic growth used in the empirical model specified below is dictated by the availability of data. Most of the data are collected from the General Statistics Office (GSO), the Ministry of Planning and Investment (MPI), the Ministry of Labour Invalids and Social Affairs (MOLISA), the Ministry of Finance (MOF) and the Ministry of Industry (MOI). See Table 2 for variable and definition and data sources.

Based on the existing literature (specifically the endogenous growth theory) and given the data availability constraints, the linkage between economic growth and financial development in Vietnam is empirically examined by making use of the following model:

$$G_{it} = \alpha_0 + \alpha_1 E_{it} + \alpha_2 HC_{it} + \alpha_3 GAP_{it} + \alpha_4 LD_{it} + \alpha_5 FDI_{it} + \alpha_6 INF_{it} + \alpha_7 FD_{it} + \alpha_8 PI_{it} + \alpha_9 EXR_{it} + \varepsilon_{it} \quad (1)$$

where G_{it} is growth of rate of the real GPP_{it} per-capita, α_i 's are the unknown population coefficients and ε_{it} is the error term.

The expected signs of the estimated coefficients are described in Table 3.

Within the context of developing countries, Fischer (1993) has argued that macroeconomic factors affect economic growth through uncertainty. For example, uncertainty associated with high inflation can reduce the growth rate of productivity

Table 2 Variable definitions and data sources

Abbreviations	Variables	Source
G	Growth rate of the real Gross Province Product (GPP) per capita (annual %)	GSO
FD	A set of variables of financial development includes the ratio of savings to GPP (S), the ratio of credit to GPP (C), and the ratio of M2 to GDP (M2)	GSO
E	Ratio of exports of each province to its GPP	GSO
HC	Number of University and College students per thousand persons at the province level	GSO
LD	Learning by doing (annual manufacturing value added as a percentage of GPP is used as a proxy)	MOI
FDI	Stock of FDI per capita at the province level (in thousands of VND at constant prices)	MPI
GAP	Technology gap is measured as the percentage difference between the average productivity of foreign and local firms.	MOLISA
INF	Annual rate of inflation at the province level	GSO
PI	Government size is measured as the ratio of public expenditure in each province to GPP	MOF
EXR	Vietnamese Real exchange rate	GSO

Saving is measured at the province level. Credit includes loans extended by banks and other intermediaries to both business and consumers at the province level. M2 is measured at the national level. While examining the role of FDI in enhancing economic growth in ASEAN-5 economies, Bende-Nabende et al. (2001) have also used manufacturing sector value added as a percentage of GDP as a proxy for learning by doing

and hence its effect on economic growth can be negative. However, the traditional growth theories tend to emphasize the positive impact of inflation on capital accumulation which arises as a result of the portfolio shift. A rising inflation rate reduces the rate of return on money which forces investors to invest in real assets that are likely to be used for further production and hence the impact of rising inflation on economic growth can be positive.

Table 3 The expected signs of the estimated coefficients

Independent variables	Dependent variable: G
E	+
HC	+
GAP	—
LD	+
PI	±
INF	±
C	+
S	+
M2	+
EXR	—
FDI	+

Recent empirical studies have used real exchange rate as an indicator of macroeconomic stability. The real exchange rate volatility is regarded as an indicator for poor macroeconomic policies that lead to real exchange rate misalignment (Husain et al. 2005; Kamin and Rogers 2000). The relationship between the real exchange rate and economic growth rate is expected to be negative.

It has been suggested that, under certain conditions, public investment in developing countries can negatively influence economic growth. For example, Durham (2004) argues that public investment financed by increasing taxes can further increase distortions in the economy and hence increase the cost of input which can have a negative impact on expected output growth. On the other hand, public investment in infrastructure and human capital development can positively affect economic growth. Blankenau and Simpon (2004) have argued that governments play an essential role in human capital accumulation by providing funds for formal schooling. Public education expenditures directly affect human capital accumulation and consequently influence long-term growth.

Since most variables considered in this paper can be regarded as endogenous, simple OLS estimation can lead to inconsistent regression coefficient estimates. In order to overcome this difficulty, the above endogenous growth equation was estimated by making use of the generalized method of moments (GMM), which can deal with the possible simultaneity between financial development and economic growth. Lagged dependent and independent variables are used as instruments.³ All standard errors of estimates are corrected with the Newey and West procedure and thereby are heteroscedasticity and autocorrelation consistent. We examine the appropriateness of the instruments with Hansen's test of the over-identifying restrictions. The instruments are appropriate if we cannot reject the null hypothesis. Besides, the Durbin-Wu-Hausman test was used to test for endogeneity. The null hypothesis was rejected, suggesting that OLS estimates might be biased and inconsistent and hence OLS was not an appropriate estimation technique.

Based on the diagnostic tests (see Tables 4 and 5), the GMM results were found to be reliable in relative terms.

4 Data analysis

Tables 4 and 5 show the relationship between different measures of financial development and economic growth in 61 provinces of Vietnam during the period 1997–2006. Table 4 suggests that the ratio of credit to GPP is an important determinant of the provincial economic growth in Vietnam. The estimated coefficient of credit (C) in Table 4 is significant at 5% level. In other words, one can argue with 95% confidence that an increase in the ratio of credit to GPP in Vietnam increases economic growth.

³ It has been argued that lagged variables do not always serve as good instruments and the estimated results may be sensitive to the choice of instruments. Accordingly, one should try to use other suitable instrumental variables. Unfortunately, lack of data prevented us from using other instrumental variables. Recent studies such as Barbosa and Eiriz (2009) and Suyanto et al. (2009) have also highlighted this problem but following other studies involving developing economies, they have also used lagged variables as instruments. We are grateful to the referee for highlighting this problem.

Table 4 GMM Estimation of Eq. 1

Independent variables	Dependent variable: economic growth (G)		
	(1) Estimation without Asian financial crisis dummy variable	(2) Estimation with interaction between FDI and financial development	(3) Estimation with Asian financial crisis dummy variable
The ratio of credit to GPP (C)	1.8321 (2.05)**	1.9390 (2.20)**	1.7831 (2.09)**
Human capital (HC)	0.1240 (3.26)*	0.1214 (3.18)*	0.0885 (2.66)*
Technology gap (GAP)	−2.1733 (−2.92)*	−2.1692 (−2.92)*	−2.2080 (−2.80)*
Learning by doing (LD)	0.0263 (3.79)*	0.0265 (4.09)*	0.0235 (3.96)*
Public expenditure (PI)	0.6566 (1.49)***	0.6989 (1.46)***	0.6306 (1.47)***
Inflation (INF)	0.0423 (0.97)	0.0423 (0.97)	0.0367 (0.93)
Foreign direct investment (FDI)	0.00003 (2.65)*	0.00003 (2.34)*	0.00003 (2.88)*
Exports (E)	0.1567 (0.95)	0.1709 (1.02)	0.1923 (1.23)
Real exchange rate (EXR)	−0.0755 (−3.67)*	−0.0757 (−3.07)*	−0.0654 (−2.55)*
FDI*C		−0.00009 (−2.22)**	
Dummy variable for Asian financial crisis			−2.1512 (−4.33)*
Constant	18.0046 (5.18)*	17.9281 (5.18)*	17.8581 (−5.74)*
Number of observations	610	610	610
Adjusted R ²	0.28	0.30	0.28
Hansen test (<i>p-value</i>)	0.69	0.64	0.67
Durbin-Wu-Hausman (<i>p-value</i>)	0.00	0.00	0.00

(i) Robust *t*-statistics in parentheses; (ii) ***significant at 10%, **Significant at 5%, and *significant at 1%

Other important determinants of economic growth in Vietnam are exports, government expenditure, learning by doing, human capital, inflation and real exchange rate. The estimated coefficient of government expenditure in Eq. 1 is positive and statistically significant at 10% level. The effect of the ratio of exports to GPP is statistically insignificant. The estimated coefficient of learning by doing is statistically

Table 5 GMM Estimation using alternative measures of financial development

Independent variables	Dependent variable: economic growth (G)		
	(1)	(2)	(3)
	Estimation without dummy variable for Asian financial crisis 1997–2006	Estimation with dummy variable for Asian financial crisis 1997–2006	Estimation with M2 variable 2000–2006
The ratio of savings to GPP (S)	0.3973 (3.24)*	0.4297 (3.58)*	
The ratio of M2 to GDP (M2)			1.0909 (2.57)*
Human capital (HC)	0.0089 (0.61)	0.0041 (0.28)	0.0104 (2.59)*
Technology gap (GAP)	−2.0668 (−3.08)*	−2.1453 (−3.03)*	−1.3475 (−2.28)**
Learning by doing (LD)	0.0084 (1.73)***	0.0075 (1.57)***	0.0196 (3.12)*
Public expenditure (PI)	0.9582 (2.95)*	0.9170 (3.07)*	−0.3468 (−2.70)*
Inflation (INF)	0.0369 (1.01)	0.0206 (0.76)	0.1055 (2.80)*
Foreign direct investment (FDI)	0.00006 (4.94)*	0.00006 (4.56)*	0.00003 (1.69)***
Exports (E)	0.1628 (1.53)***	0.1549 (1.57)***	0.00001 (0.49)
Real exchange rate (EXR)	−0.0843 (−3.81)*	−0.0675 (−3.20)*	−0.0623 (−2.72)*
Dummy for the Asian financial crisis		−1.9855 (−4.37)*	
Constant	17.5463 (5.57)*	17.4711 (6.25)*	6.5159 (2.15)**
Number of Observations	610	610	427
Adjusted R ²	0.37	0.40	0.42
Hansen Test (<i>p-value</i>)	0.23	0.24	0.27
Durbin-Wu-Hausman (<i>p-value</i>)	0.00	0.00	0.00

The data on M2 was not available for the period 1997–1999. As a result of this, we were unable to estimate the relationship between M2 and economic growth before 2000

(i) Robust *t*-statistics in parentheses; (ii) ***significant at 10%, **Significant at 5%, and *significant at 1%

significant at 1% level and its sign is consistent with expectations. The significance of learning by doing perhaps reflects the fact that some production activities involve assembling only. In addition, the Vietnamese labour force is benefiting from knowledge spillovers thereby improving its productivity and hence stimulating

economic growth. The estimated coefficient of the real exchange rate has an expected negative sign and it is significant at 1% level. The impact of FDI on economic growth is positive and statistically significant at 1% level. The impact of human capital on economic growth is statistically significant at 1% level and the sign is positive.

Recent empirical studies have argued that the impact of FDI on economic growth also depends on the existence of adequate absorptive capacity in host economies. Absorptive capacity of an economy can be measured by factors such as the stock of human capital, the level of financial market development and the extent of technology gap between the foreign and local firms. It has been argued that FDI has a direct and an indirect effect on economic growth. The direct effect arises from FDI-led increase in the supply of capital which increases the overall production capacity of the host economy. The indirect effect on economic growth arises from FDI's interaction with factors such as the level of financial development, the stock of human capital and the extent of technology gap. In order to examine the effect of FDI on economic growth through Vietnam's absorptive capacity, Eq. 1 is re-estimated after introducing the interaction of FDI and the Level of Financial Market Development as additional independent variables.

As can be seen in column 2 of Table 4, the estimated coefficient of the interaction between FDI and the level of financial development (C) is negative and statistically significant at 5% level. This suggests that, as far as the level of financial market development is concerned, Vietnam has not reached the minimum required threshold. In other words, by further developing its financial markets, Vietnam can take more advantage of FDI inflows.

Vietnam's economy was significantly affected by the Asian financial crisis. Thus, we extended our model by introducing a one-time dummy variable for the Asian financial crisis in 1997.⁴ Column 3 of Table 4 shows the estimated results when Eq. 1 was re-estimated by GMM after inclusion of a financial crisis dummy variable. The financial crisis dummy variable has a negative sign in the economic growth equation. In economic growth equation, it is statistically significant at 1% level. The introduction of the dummy variable leads to a minor change in the magnitude of the estimated coefficients without affecting their significance level. The estimated results suggest that the Asian financial crisis negatively affected economic growth in Vietnam.

We used different measures of financial development such as the ratio of saving to GPP and the ratio of M2 to GDP. Table 5 shows that, as expected, all coefficients of the ratio of savings to GPP and the ratio of M2 to GDP are positive and statistically significant at the 1% level. The estimated results imply that both gross domestic savings and the money aggregate contributed to economic growth. Other important determinants of economic growth in Vietnam are exports, government expenditure, learning by doing, human capital, inflation and real exchange rate.

5 Conclusions

The causal relationship between financial development and economic growth remains an issue of intense debate among researchers. While this debate has

⁴ The value of the dummy variable was set equal to 1 in 1997 and 0 in all other years.

provided rich insights into the relationship between financial development and economic growth in a number of developing countries, because of the lack of data, few empirical studies have considered the case of Vietnam.

By making use of a recently released panel dataset which covers 61 provinces of Vietnam over the period 1997–2006, this paper attempts to examine the link between financial development and economic growth and related issues. The empirical analysis, which is based on endogenous growth model, suggests that financial development has facilitated economic growth in Vietnam. We found a positive relationship between the ratio of credit to GPP and economic growth in Vietnam. This conclusion was also supported when alternative measures of financial development such as gross domestic saving and monetary aggregate M2 were used. We also considered the impact of the interaction between the level of financial development as measured by the ratio of credit to GPP and the stock of foreign direct investment on Vietnam's economic growth rate. Our empirical analysis found that the interaction has a negative and statistically significant effect on Vietnam's economic growth rate. This suggests that, as far as the level of financial market development is concerned, Vietnam has not reached the required minimum threshold. In other words, by further developing its financial market, Vietnam can take greater advantage of the stock of foreign direct investment. Despite registering rapid economic growth over a significant period of time, Vietnam's financial market remains relatively underdeveloped. The positive impact of foreign direct investment on economic growth in Vietnam will be larger if more resources are invested in financial market development. At present, Vietnam's financial market is dominated by its banking sector. Therefore, diversification of the financial services sector is highly desirable.

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